

To the Editors
Scientific Reports

Re: Article submission — “Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis.”

Dear Editors,

We are pleased to submit the above manuscript for consideration as an Article in *Scientific Reports*. Using a pre-registered specification-curve design, we measure whether large-language-model credit ratings are stable when the underlying financial information is held fixed but the elicitation specification varies. Across a fixed 90-item battery crossed with a factorial grid of provider, model version, temperature, prompt paraphrase, output format, few-shot exemplars, and presentation (12,960 elicitations across three model families), no controllable elicitation axis explains more than about 5% of rating variance, the instability survives verified deterministic execution (permutation $p = 0.001$), and 25.2% of matched elicitation pairs disagree at the investment-grade/high-yield boundary. We translate this into decision-relevant units — 53.5% implied portfolio turnover and a \$2.0–4.1bn Pillar-1 capital range on the same \$45bn book — and show the pattern persists on 31 decontaminated real issuers under a pre-registered fingerprinting gate. Our estimand throughout is rating stability, not accuracy.

The work fits *Scientific Reports* through its broad relevance to the reliability of AI decision systems and its methodological rigor: the design is pre-registered, inference is distribution-free, and a deterministic, offline reproducibility capsule archived on Zenodo (<https://doi.org/10.5281/zenodo.21953935>; MIT license) regenerates every reported number, table, and figure byte-for-byte. The study uses only synthetic firm profiles and publicly available SEC filings, involves no human or animal subjects, and discloses its use of generative-AI assistance in the Methods.

The manuscript is original, is not under consideration elsewhere, and has been approved by both authors, who declare no competing interests and received no external funding. We have not discussed this work with a *Scientific Reports* Editorial Board Member and request no referee exclusions. Thank you for your consideration.

Sincerely,

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